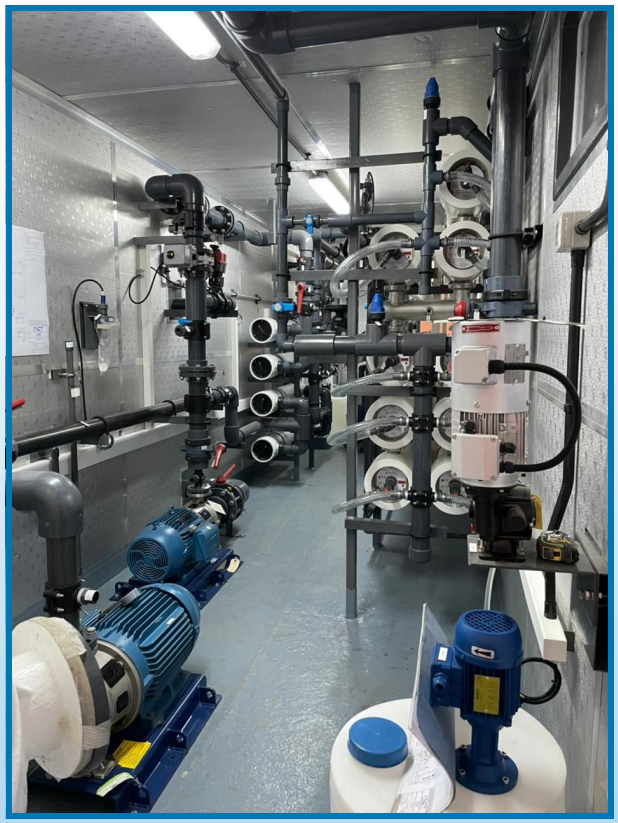
WHITEPAPER

FEDCO

Making fluid energy work for you™






# BiTurbo™

## More Water at the Lowest Cost: A Revolution in RO Technology

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# BiTurbo™

## A Revolution in High Recovery RO

Customize recovery - up to 60% for SWRO at standard pressure

Optimize membrane performance

Optimize permeate production

Reduce total cost of water

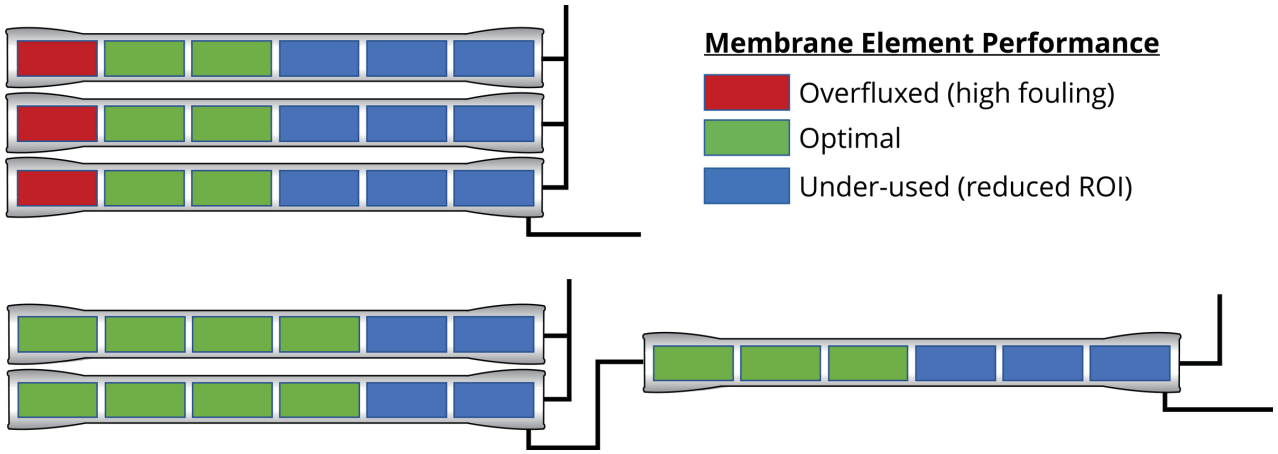
Minimize environmental impact

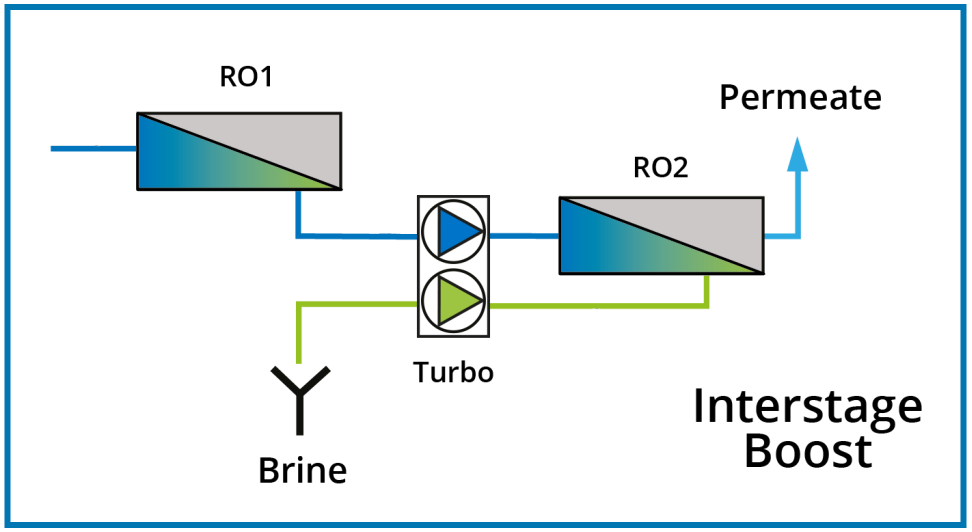
Validated by major membrane manufacturers

# Multi-stage RO:
# Improving recovery and performance

Multi-stage RO allows **flux balancing** across a membrane array.

This delivers better membrane performance, **higher recovery and more stable operation** than single-stage systems.



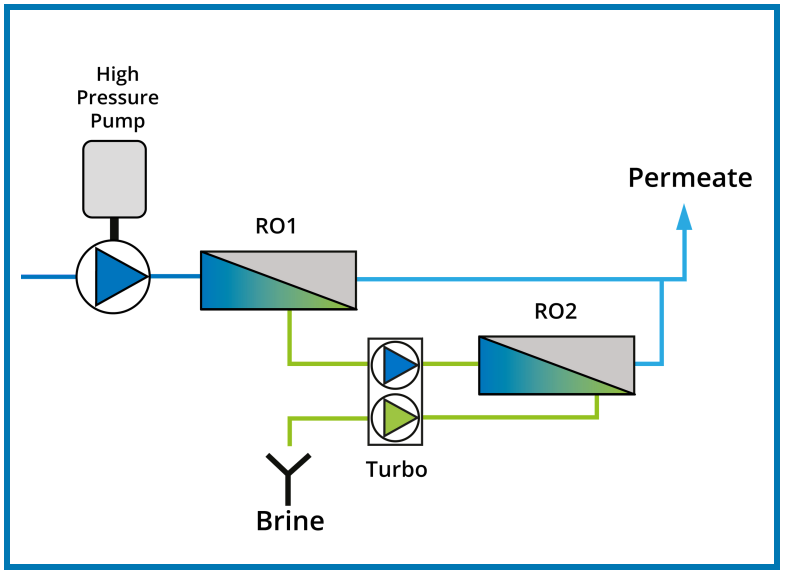

Multi-stage RO is commonplace in brackish water desalination. The crucial component to achieving well balanced flux rates is the addition of an **interstage booster**.

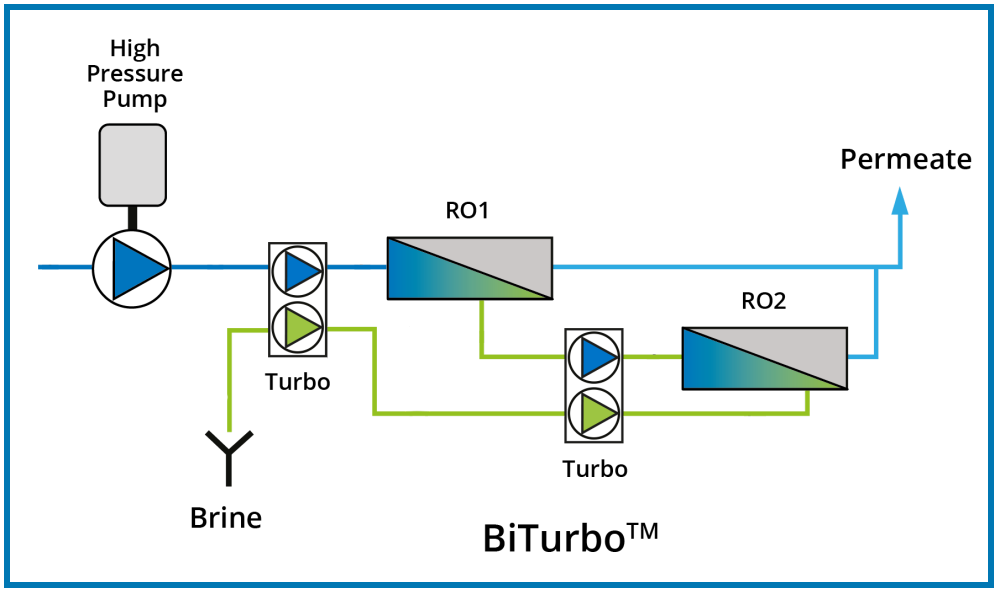
This can best achieved by using a **turbocharger powered by brine energy**.

# Applying multi-stage RO to seawater desalination

*Using interstage boosting to improve recovery and flux balancing has been proven to be effective in seawater and brackish water reverse osmosis applications.. The **BiTurbo**™ uses highly efficient turbochargers to apply this technique to seawater desalination.*

In a **brackish water system**, a 3 bar interstage boost requires almost all available brine energy due to high recovery and low operating pressure.




In contrast, for **seawater desalination** even a 15 bar interstage boost requires just a fraction of available brine energy.

The remaining energy is harnessed by a feed turbocharger.

# Implementing the BiTurbo™

*Multi-stage RO offers system designers maximum flexibility:*

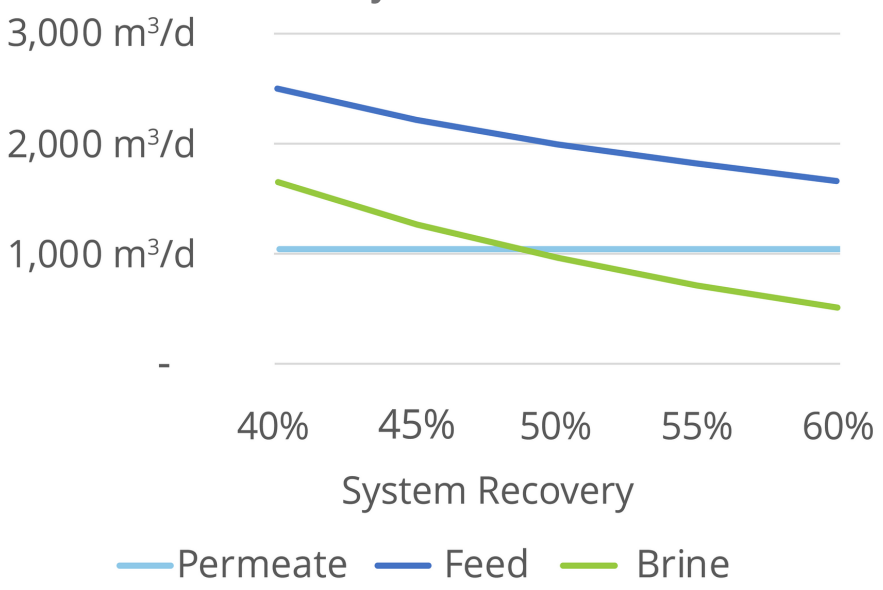
- *Increase permeate production without increasing the scope of civil works*
- *Keep permeate production constant and decrease the scope of civil works*

Where demand for desalinated water is known, increasing RO recovery brings down the total cost of water by reducing the scope of civil works

Where demand for water is increasing, or where available feed water is limited by well capacity or permitting, increasing RO recovery allows for higher system capacity without expanding the scope of civil works

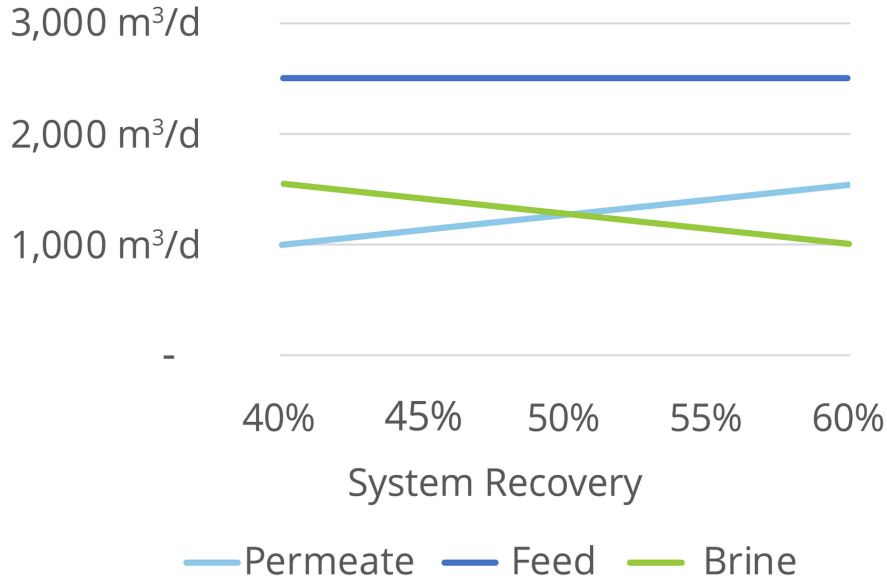
## Scenario 1

***Keep permeate production constant, decrease scope of civil works***




## Scenario 2

***Increase permeate production, scope of civil works remains the same***



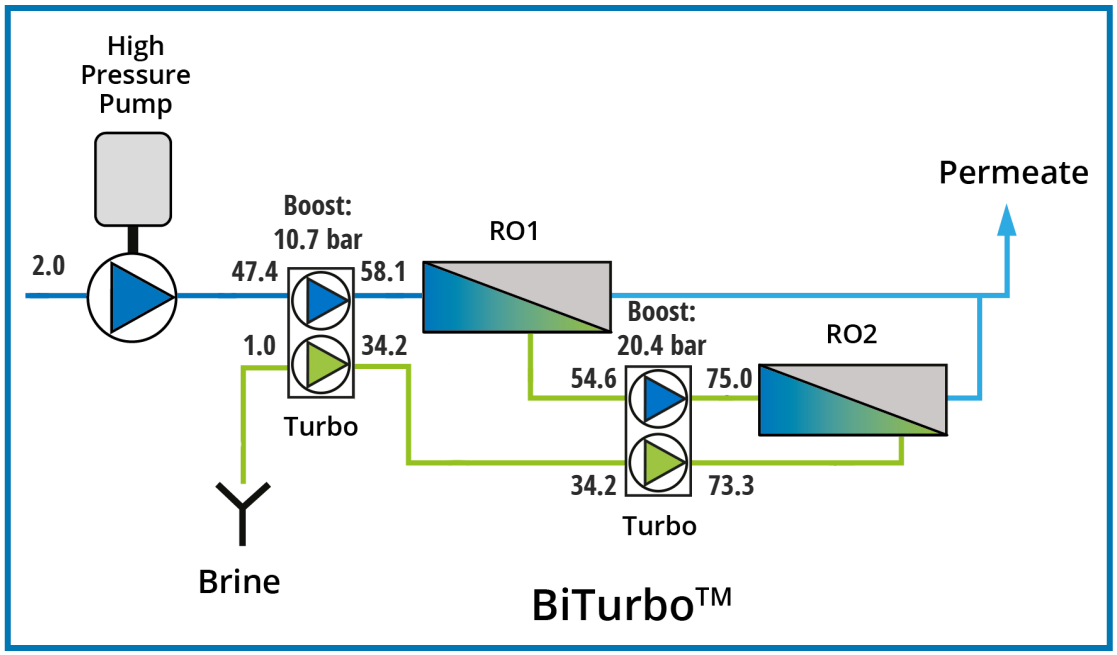
# Using turbochargers for interstage boosting

Turbochargers can provide i**nterstage pressure boosting**, by using a brine-powered turbine to drive an impeller. This improves membrane performance by balancing flux rates across the RO system.

In a **BiTurbo™** system, Feed and interstage pressure boosts are driven entirely by brine energy – no additional motors or drives are required. The system needs only three (3) rotating parts: one high pressure pump and two turbochargers.



*The world's largest turbochargers handle feed flows of up to 3,460 $m^3$/hr*




**Flexible and reliable operation**

This example BiTurbo™ system has a feed TDS of 36,000 ppm, at **57% recovery**. Pressure in bar is indicated at each point in the system

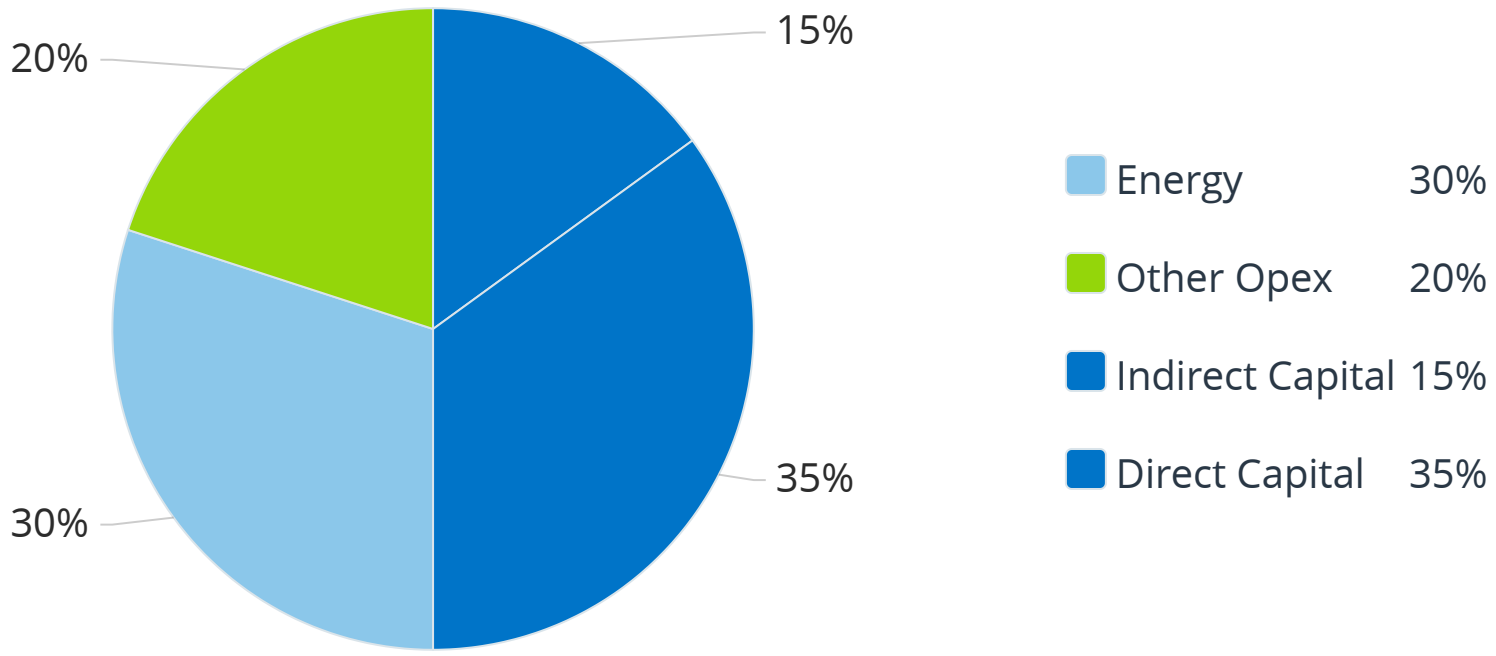
The system's second-stage operates at 75 bar and a maximum brine TDS of 84,000 ppm, allowing reliable operation wth standard SWRO equipment. The system is simple to adjust to fluctuations in feed salinity.

# Cost of water: why does reducing capex matter?

The desalination industry has focused on energy consumption for the past two decades. However, CAPEX and interest payments represent **30-50% of the total cost of water** of a desalination project.

**Cost Breakdown of a Typical SWRO Project**

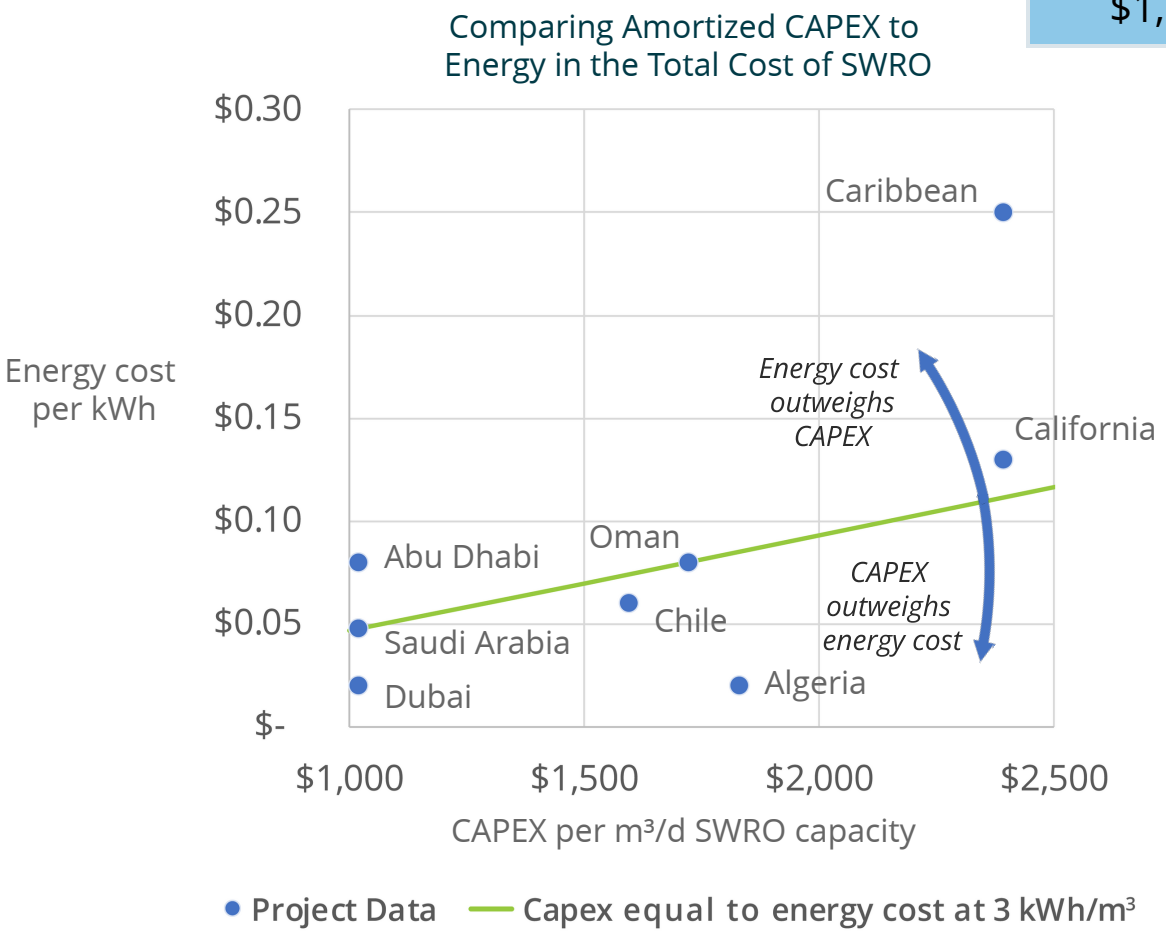
Voutchkov, 2018



# Considering the lifetime cost of CAPEX

The rule of thumb of $1,000 per m³/d of SWRO capacity expresses **direct CAPEX**. 80% of this is generally financed, meaning the **cost of interest** is a vital consideration.

| Cost of Finance | | | | |
|---|---|---|---|---|
| **Unit cost ($/m³/d)** | **3%** | **5%** | **7%** | **9%** |
| $1,000 | $1,275 | $1,539 | $1,818 | $2,116 |
| $1,250 | $1,594 | $1,924 | $2,273 | $2,645 |
| $1,500 | $1,913 | $2,308 | $2,727 | $3,175 |




This chart maps approximate capital and energy costs in different locations.

The further below the dashed line a point is, the greater the proportion of total water cost is represented by CAPEX.

In many locations CAPEX represents an equal or greater proportion of total cost of SWRO in comparison to energy costs.

# BiTurbo™ references

**The BiTurbo™ technology has been tried and tested on three continents in a variety of feed water conditions, with many more installations under development.**

Clients and end-users in Mexico, Saudi Arabia and Indonesia rely on the BiTurbo™ for high SWRO recovery, low energy consumption and highly reliable operation. The small footprint of a BiTurbo™ system also provides flexibility by leaving room for future expansions to capacity.



A BiTurbo™ system operating at 55-57% recovery in Los Cabos, Mexico

In addition to references in the field, our in-house, closed-loop BiTurbo™ pilot system enables us to partner with key global membrane suppliers in ongoing research on high-recovery seawater desalination.

| Reference Location | Capacity ($m^3/d$) | Online Date | OEM | End User | Sector |
|---|---|---|---|---|---|
| Los Cabos, Mexico | 360 | Feb 2019 | Water Technologies de Mexico | Solmar, Rancho San Lucas | Tourism |
| Indonesia (2 systems) | 3,000<br>3,000 | Dec 2019<br>July 2023 | Confidential | Confidential | Tourism |
| Jubail, Saudi Arabia | 672 | Feb 2022 | PACT Engineering | SWCC/DTRI | Brine Mining Pilot |
| Cancun, Mexico | 700 | Feb 2022 | Sonitec | Fiesta Americana Hotel | Tourism |
| San Quintin, Mexico | 2,300 | May 2022 | World Water | BerryMex | Agriculture |
| Los Cabos, Mexico | 5,400 | Sept 2023 | TSG Water | Confidential | Tourism |

# High recovery RO minimizes environmental impact

The majority of a desalination plant's physical impact on the environment is a function of feed flow. With a lower feed flow for the same permeate production, high recovery RO reduces environmental impact across the plant's life cycle.



Smaller intake structures reduce impact of marine construction



Smaller plant footprint reduces physical impact of desalination

High recovery brings down the carbon footprint of desalination by reducing energy consumption, chemical use, and equipment size.

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# BiTurbo™

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Contact our team to find out more: sales@fedco-usa.com